

PCOS NUTRITION GUIDELINES

Hormone-Friendly, Insulin-Sensitive Dietary Reference

1. What is PCOS?

Polycystic Ovary Syndrome (PCOS) is the most common endocrine disorder in women of reproductive age, affecting 6–12% of women globally. It is defined by the Rotterdam Criteria (2 of 3): oligo/anovulation, clinical/biochemical hyperandrogenism, and polycystic ovaries on ultrasound.

Insulin Resistance is present in 65–80% of PCOS patients regardless of BMI. Hyperinsulinemia stimulates ovarian androgen production, worsening hormonal imbalance, acne, hirsutism, and menstrual irregularity. Addressing insulin resistance through diet is a cornerstone of PCOS management.

2. Dietary Goals

- **Improve Insulin Sensitivity:** Low-GI diet, balanced macros, fiber-rich foods reduce fasting insulin by 20–30%.
- **Reduce Chronic Inflammation:** PCOS is an inflammatory condition. Anti-inflammatory foods (omega-3, antioxidants) reduce CRP and inflammatory cytokines.
- **Support Hormonal Balance:** Adequate zinc, magnesium, and vitamin D regulate androgen metabolism and SHBG levels.
- **Achieve Healthy Weight:** Even 5–10% weight loss in overweight PCOS patients restores ovulation in 50–60% of cases.
- **Gut Microbiome Support:** Prebiotic fiber and fermented foods improve estrogen metabolism and reduce systemic inflammation.

3. Carbohydrate Management

Target carbohydrates: **40–45% of total energy**. Emphasize low-GI, high-fiber choices. Avoid refined carbohydrates as they acutely elevate insulin, worsening androgen excess.

Category	Preferred Choices	Choices to Limit/Avoid
Grains	Oats, quinoa, barley, broken wheat (dalia), millets	White rice, maida, suji, instant noodles
Legumes	All lentils, chickpeas, kidney beans, edamame	Canned beans in sugar sauces
Vegetables	All non-starchy: broccoli, spinach, cauliflower, bell peppers	Potato (large quantities), corn (in excess)
Fruits	Berries, apple, pear, guava, pomegranate (small portions)	Mango, banana, chiku, fruit juices, dried fruits in excess

4. Protein Intake

Recommended: 1.2–1.8 g/kg body weight/day. Higher protein intake improves satiety, supports lean mass, reduces total caloric intake, and lowers postprandial insulin response.

Source	Amount	Protein (g)	Notes
Paneer (low-fat)	100g	~18g	Good calcium + protein; limit high-fat variety
Lentils (cooked)	1 cup (200g)	~18g	Also provides fiber and iron
Eggs	2 whole	~12g	Complete amino acid profile
Chicken breast (grilled)	100g	~27g	Lean protein, avoid fried preparation
Greek yogurt	150g	~15g	Probiotic benefits; choose unsweetened
Tofu	100g	~8g	Phytoestrogen content — moderate use
Mixed seeds (flax+chia)	2 tbsp (20g)	~5g	Also omega-3, fiber

5. Healthy Fats — Omega-3 Priority

Omega-3 fatty acids (EPA/DHA) reduce circulating testosterone, improve menstrual regularity, and lower triglycerides in PCOS (RCT evidence). Target: **2–3 servings of omega-3-rich foods per week.**

- Fatty fish: salmon, mackerel, sardines, hilsa (2–3×/week)
- Walnuts: 2–3 halves daily (ALA source)
- Flaxseed (ground): 1–2 tbsp/day
- Chia seeds: 1–2 tbsp/day
- Algae-based omega-3 supplement (for strict vegetarians)

6. Anti-Inflammatory Foods

Food / Compound	Active Component	Benefit in PCOS	How to Use
Turmeric	Curcumin	Reduces IL-6, TNF-alpha; improves insulin sensitivity	1/2 tsp in warm milk or cooking daily
Leafy greens (spinach, methi)	Magnesium, folate, antioxidants	Reduces oxidative stress; supports ovulation	1–2 cups cooked daily
Berries (amla, blueberry)	Polyphenols, Vitamin C	Lowers CRP, improves lipid profile	1/2 cup fresh or amla juice daily
Ginger	Gingerols, shogaols	Anti-inflammatory; reduces menstrual pain	Fresh ginger in tea or meals
Cinnamon	Cinnamaldehyde	Improves insulin receptor sensitivity	1/4–1/2 tsp in oats, smoothies
Green tea	EGCG (catechins)	Reduces androgen levels, aids weight loss	2–3 cups/day; no added sugar
Dark chocolate (>70%)	Flavanoids	Reduces cortisol, improves insulin sensitivity	1–2 small squares; occasional treat

7. Foods to Avoid

- **Sugar & sugary beverages:** Acutely spikes insulin — worsen androgen production.
- **Refined carbohydrates:** White bread, maida, pastries — high GI, inflammatory.
- **Trans fats:** Vanaspati, partially hydrogenated oils — increase inflammation and insulin resistance.
- **Excessive dairy (full-fat):** May elevate IGF-1 and androgen activity in susceptible women.
- **Alcohol:** Impairs liver estrogen metabolism; increases cortisol.
- **Processed/packaged foods:** High sodium, preservatives, hidden sugars.
- **Excessive soy:** High isoflavone content may disrupt hormonal balance when consumed in large amounts.

8. Key Micronutrients in PCOS

Nutrient	Role in PCOS	Deficiency Risk	Food Sources
Magnesium	Improves insulin receptor signaling; reduces cortisol	HIGH — 80% deficient	Dark leafy greens, pumpkin seeds, dark chocolate, banana
Zinc	Regulates androgen metabolism; reduces acne; supports ovulation	MODERATE	Pumpkin seeds, sesame, legumes, eggs, lean meat
Vitamin D	Regulates insulin secretion, AMH, and ovarian function	VERY HIGH — up to 85%	Sunlight, fatty fish, fortified dairy, egg yolk; supplement often needed
Inositol (Myo+D-chiro)	Acts as insulin second messenger; improves egg quality	Functionally depleted	Citrus peel, nuts, whole grains; supplement 2–4g/day (evidence-based)
Iron	Often low due to heavy menstrual bleeding	MODERATE	Lentils, spinach (with Vit C), lean meat, fortified cereals
Folate/B9	Supports ovulation, fertility	LOW-MODERATE	Leafy greens, beans, fortified foods

9. Sample PCOS-Friendly Meal Plan (1600 kcal)

Meal	Time	Items	Key Benefits
Early Morning	7:00 AM	Warm water + 1 tsp cinnamon + lemon; 5 soaked almonds	Insulin sensitizing start
Breakfast	8:30 AM	Oats porridge (50g) with chia seeds, berries, and low-fat milk	Low GI, omega-3, fiber
Mid-Morning	11:00 AM	Buttermilk (200 ml) + 1 guava or amla	Probiotics + Vit C
Lunch	1:00 PM	Quinoa/brown rice (1 cup) + mixed dal + sauteed spinach + salad + curd	Complete amino acids + fiber
Evening	4:30 PM	Roasted flaxseed + pumpkin seed trail mix (25g) + green tea	Omega-3, zinc, magnesium

Meal	Time	Items	Key Benefits
Dinner	7:30 PM	2 bajra roti + palak paneer (moderate) + vegetable soup	Iron, calcium, protein

10. Lifestyle Integration

- Exercise:** 150 min/week moderate aerobic activity (brisk walk, cycling) + 2×/week strength training. Resistance training improves insulin sensitivity by 25–30% in PCOS.
- Sleep:** 7–9 hours/night. Poor sleep worsens cortisol, ghrelin (hunger hormone), and insulin resistance. Sleep apnea is more common in PCOS — screen if indicated.
- Stress Management:** Chronic stress elevates cortisol and androgens. Yoga, mindfulness, and breathing exercises (pranayama) have RCT evidence for improving PCOS hormonal markers.
- Meal Timing:** Front-load calories (larger breakfast, smaller dinner). Late-night eating worsens circadian insulin resistance.

References: Endocrine Society Clinical Practice Guidelines 2023 • Thessaloniki ESHRE/ASRM PCOS Consensus • AJCN PCOS Nutrition Reviews 2021 • ICMR Guidelines 2020